FABER-CASTELL
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Lecture -34
PCA

- with p vars. the no of pairwise scatterplot will be $\frac{p(p-1)}{2}$.
- a direc^n in R^p is a vector ϕ of length $= 1 = \|\phi\|$
  $z = \phi^T x = \phi_1 x_1 + \phi_2 x_2 + \phi_3 x_3 \dots$
  1 number instead of p.
- Let $X \in R^{n \times p}$ be centered rows $x_1 \dots x_n$

cov. matrix $\leftarrow S = \frac{1}{n-1} X^T X \in R^{p \times p}$ $\quad S_{jk} = \hat{Cov}(x_j, x_k)$

The projected values are $z_i = \phi^T x_i$ & var.
$\text{Var}(\phi^T x) = \phi^T S \phi$

$$\boxed{\phi_1 = \arg\max_{\|\phi\| = 1} \phi^T S \phi}$$

$$L(\phi, \lambda) = \phi^T S \phi - \lambda(\phi^T \phi - 1)$$
$$\frac{\partial L}{\partial \phi} = 2S\phi - 2\lambda\phi = 0 \Rightarrow \boxed{S\phi = \lambda\phi}$$

$$\boxed{\phi^T S \phi = \lambda \phi^T \phi = \lambda}$$

Eginvalue is the var. we are maximizing. Take eginvector with highest eginvalue. This is your 1st principal component, & λ_1 is the var.

The second one must be orthogonal.

$$S = P \Lambda P^T \qquad P = [\phi_1 \dots \phi_p]$$
$$\lambda = \text{diag}(\lambda_1 \geq \dots \geq \lambda_p \geq 0)$$

- Total var. of the data is $\text{tr}(S) = \sum_j \lambda_j$
  proportion captured by 1st k comp. is

$$\frac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_p}$$

- scaling / standardization is must.
- $X = UDV^T$

$$S = \frac{1}{n-1} X^T X = \frac{1}{n-1} VD^2V^T$$

$$\Rightarrow \phi_k = V_k, \quad \lambda_k = \frac{d_k^2}{n-1}$$

- PCA → in which direc^n do the predictors vary most
  PLS → in which direc^n do the predictors vary in a way that tracks y.

	maximize	use y
PCA/PCR	$\text{Var}(X\phi)$	no
PLS	$\widehat{\text{cov}}(X\phi, y)^2$	yes

- Kernel PCA → bend the space, then run PCA.
  t-SNE → place pts. in 2D, with nearer neighbors
  UMAP → faster, keeps more of a large layout.
  Autoencoders → nonlinear compression with NN.